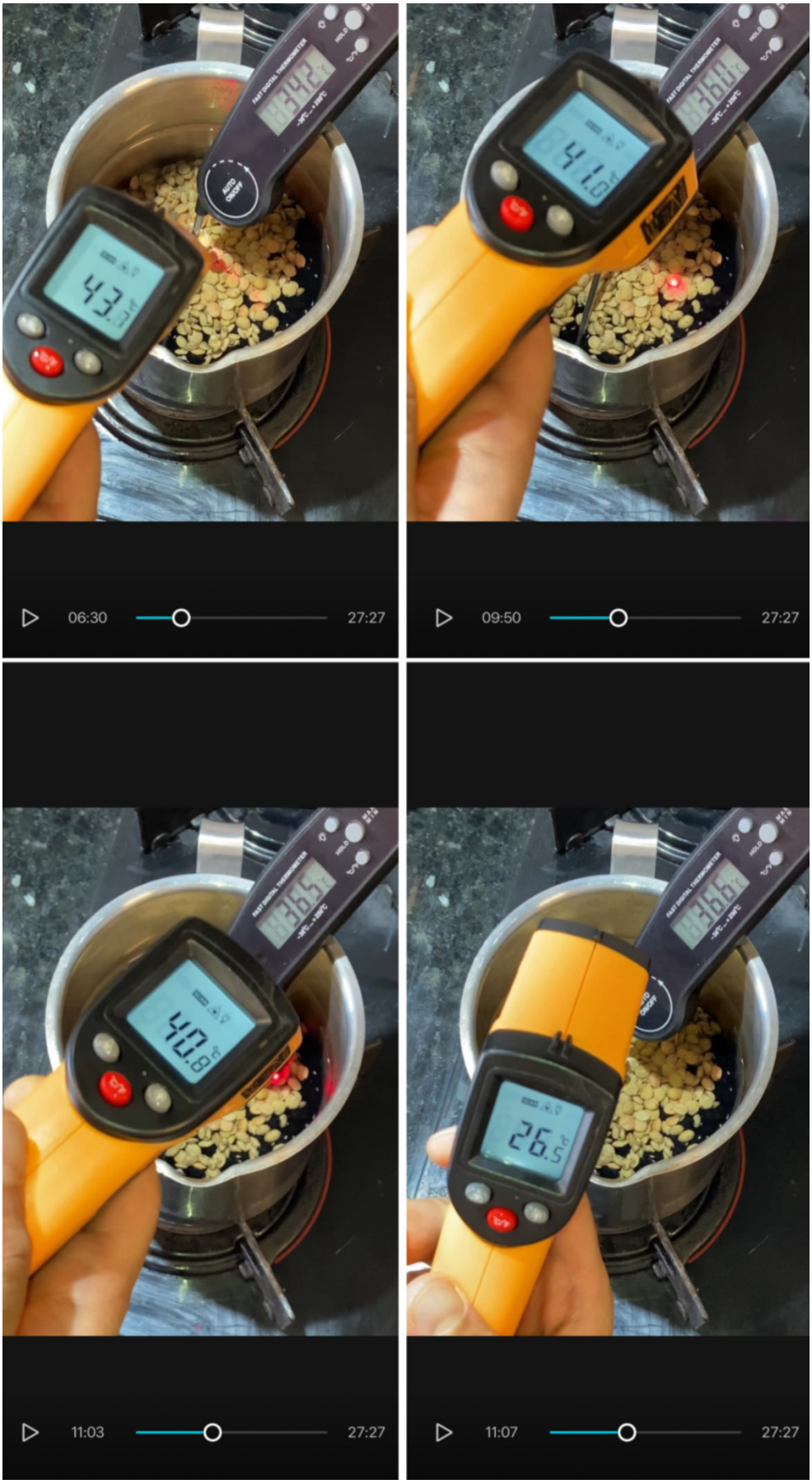

Figure 27: E2c8 — THERMAL RESERVOIR + IR CONTROL. All frames: flame OFF. [F1, 6:30, probe 34.2°C, IR 43.7°C]: Δ = 9.5°C. Surface far hotter than bottom, 4 min after flame OFF. Temperature has risen 7.8°C without any heat source. [F2, 9:50, probe 36.0°C, IR 41.0°C]: Δ = 5.0°C. Still rising, gradient converging. [F3, 11:03, probe 36.5°C, IR 40.8°C]: Δ = 4.3°C. [F4, 11:07, probe 36.6°C, IR 26.5°C — WALL]: IR pointed at kitchen wall reads 26.5°C — ambient temperature. CONTROL: Confirms IR thermometer is functioning correctly. Surface of liquid reads 40+°C, kitchen wall reads 26.5°C. The surface temperature measurement is real, not an artifact.

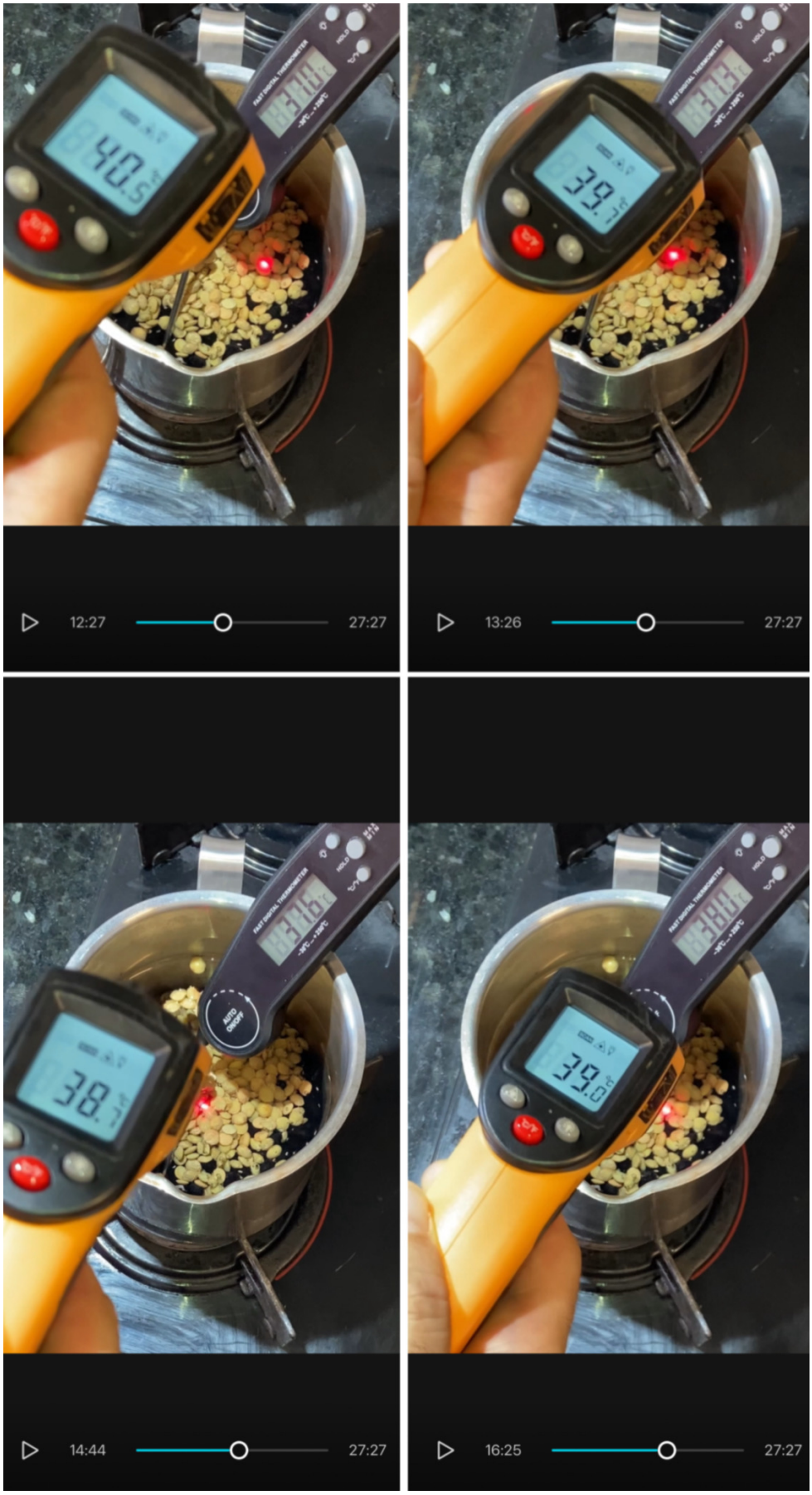

Figure 28: E2c9 — GRADIENT CONVERGENCE CONTINUES. All frames: flame OFF. [F1, 12:27, probe 37.0°C, IR 40.5°C]: Δ = 3.5°C. [F2, 13:26, probe 37.3°C, IR 39.7°C]: Δ = 2.4°C. [F3, 14:44, probe 37.6°C, IR 38.7°C]: Δ = 1.1°C. Approaching convergence. [F4, 16:25, probe 38.0°C, IR 39.0°C]: Δ = 1.0°C. Probe still rising (38.0°C), 13 min 51 s after flame OFF.

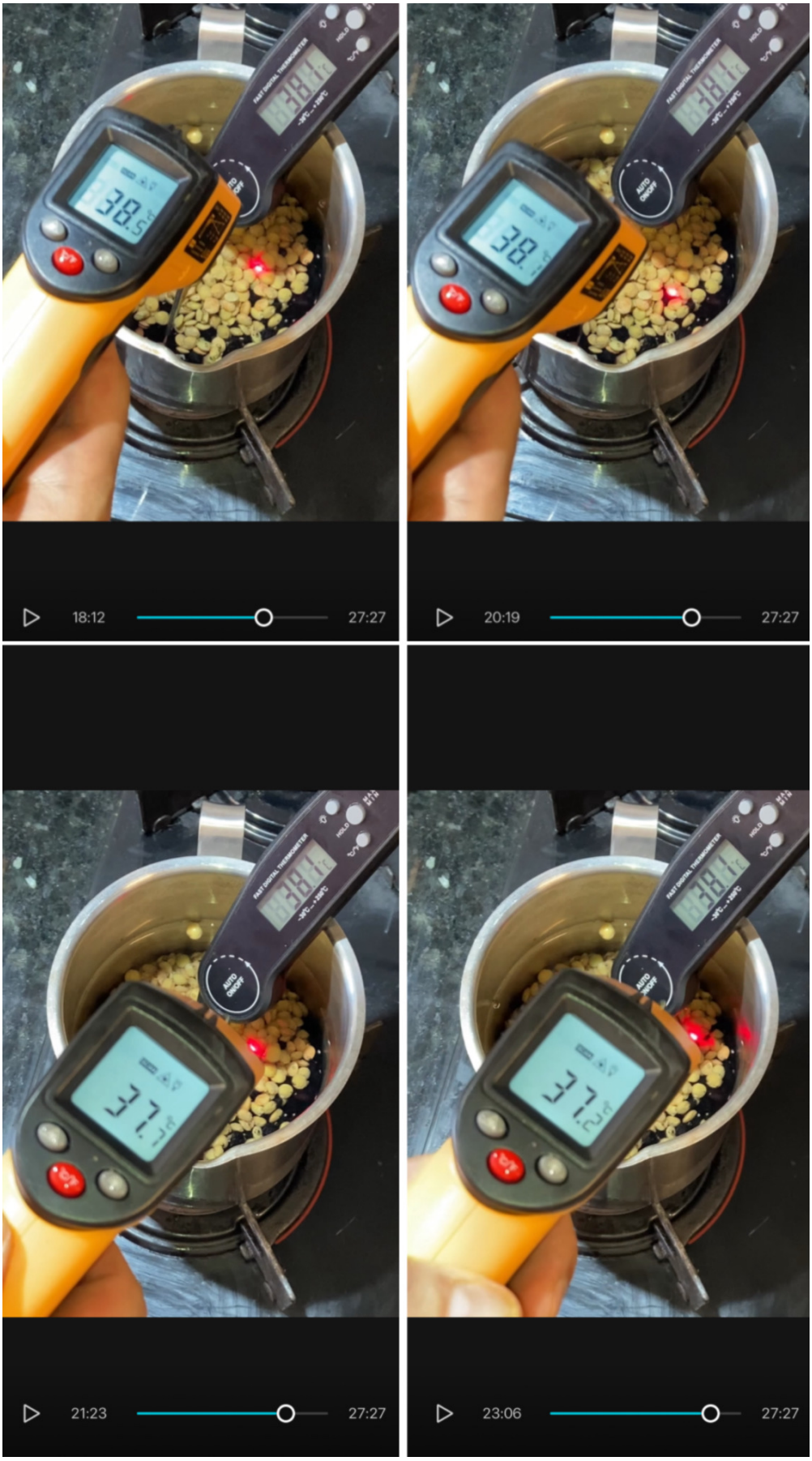

Figure 29: E2c10 — CONVERGENCE AND INVERSION. All frames: flame OFF. [F1, 18:12, probe 38.1°C, IR 38.5°C]: Δ = 0.4°C. Near convergence. [F2, 20:19, probe 38.1°C, IR 38.1°C]: EXACT CONVERGENCE. Surface = bottom = 38.1°C. This marks 17 min 45 s since flame OFF. Temperature rose from 26.4°C to 38.1°C (+11.7°C) without flame. [F3, 21:23, probe 38.1°C, IR 37.7°C]: GRADIENT INVERTED — surface now COLDER than bottom. [F4, 23:06, probe 38.1°C, IR 37.2°C]: Inversion continues. System now cooling from surface first.

Phase 6: Cooling and Second Flame Cycle (t = 24:02 – 27:10)

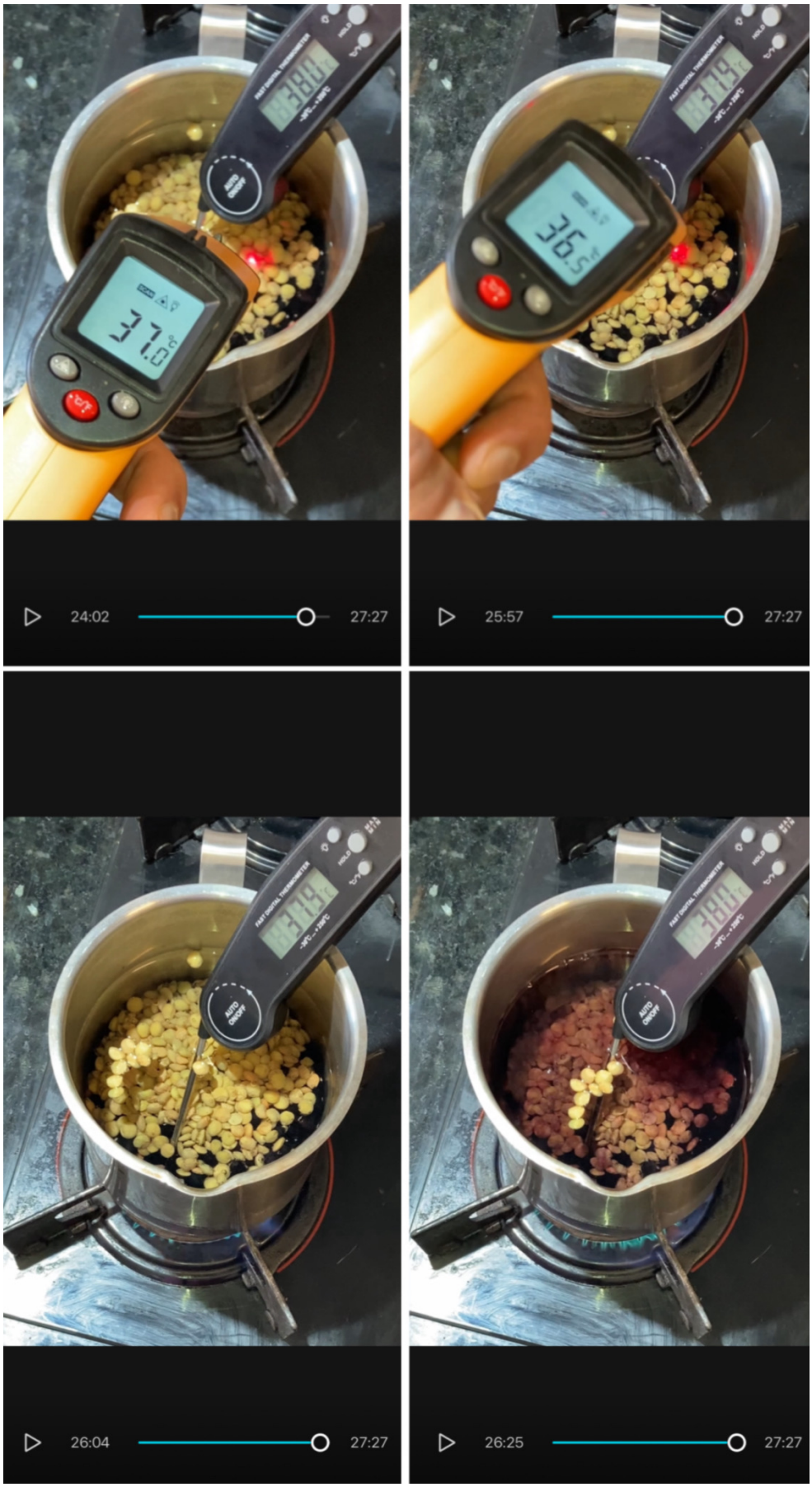

Figure 30: E2c11 — COOLING + SECOND FLAME IGNITION. [F1, 24:02, probe 38.0°C, IR 37.0°C]: Cooling continues. Δ = –1.0°C (surface colder). [F2, 25:57, probe 37.9°C, IR 36.5°C]: Δ = –1.4°C. [F3, 26:04, probe 37.9°C]: FLAME ON — second heating cycle begins. [F4, 26:25, probe 38.0°C]: 21 seconds of flame, temperature barely changed (+0.1°C).

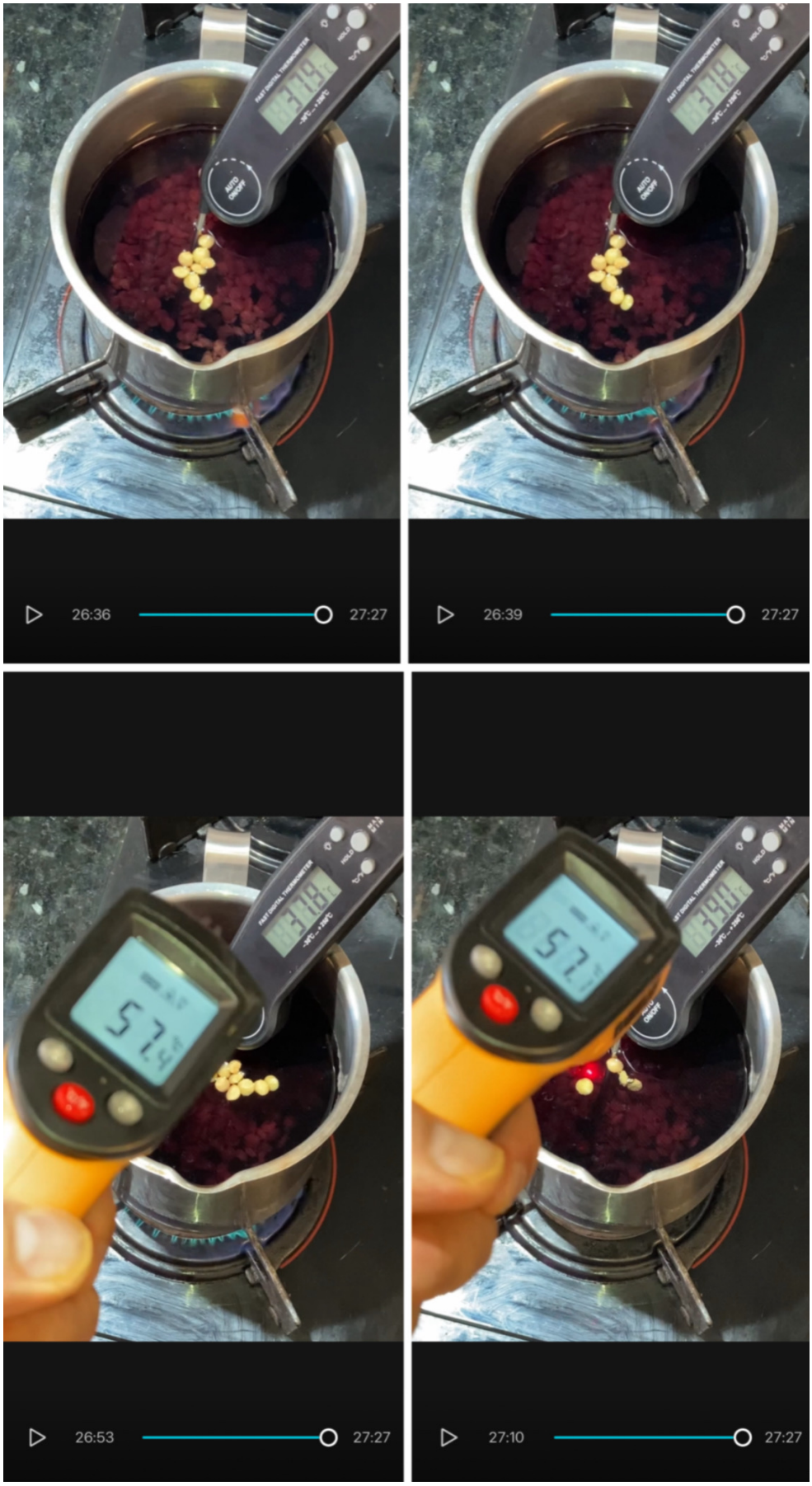

Figure 31: E2c12 — SECOND CYCLE: SURFACE EXPLOSION + END. [F1, 26:36, probe 37.9°C]: 32 s of flame, probe essentially unchanged. [F2, 26:39, probe 37.8°C]: Probe DROPS 0.1°C — temperature decrease with flame ON, reproducing Experiment 5 pattern. [F3, 26:53, probe 37.8°C, IR 57.4°C]: SURFACE EXPLOSION — IR reads 57.4°C while probe at bottom reads 37.8°C. Δ = 19.6°C in under 48 seconds of flame. The surface absorbed and concentrated the thermal energy while the bottom remains cold — consistent with the attraction→torque→repulsion cycle. [F4, 27:10, probe 39.0°C, IR 57.1°C]: Flame OFF. Δ = 18.1°C. Experiment ends. FINAL STATE:

After 63 seconds of flame, probe rose only 1.1°C (37.9→39.0) while surface jumped 20.9°C (36.5→57.4).

Complete Data: Experiment 6 (48 Frames, 12 Collages)

#	Time	Flame	Probe °C	IR °C	Δ °C	Phase
1	0:04	OFF	—	23.0	—	Baseline
2	0:18	OFF	25.6	—	—	Probe stabilizing
3	0:31	OFF	22.3	—	—	Baseline stable
4	0:54	OFF	22.3	—	—	Baseline confirmed
5	0:55	ON*	17.3	—	—	Instantaneous drop (AI)
6	0:55	ON	17.3	—	—	Drop confirmed
7	0:55	ON	20.8	—	—	Recovery start
8	0:56	ON	20.9	—	—	Recovery
9	0:57	ON	21.0	—	—	Recovery
10	0:58	ON	21.1	—	—	Recovery
11	0:59	ON	21.3	—	—	Recovery
12	1:00	ON	21.5	—	—	Recovery
13	1:01	ON	21.6	—	—	Recovery
14	1:02	ON	21.7	—	—	Recovery
15	1:02	ON	21.8	—	—	Recovery
16	1:03	ON	21.9	—	—	Recovery
17	1:04	ON	22.0	—	—	Plateau start
18	1:05	ON	22.1	—	—	Brief peak
19	1:19	ON	22.0	—	—	Plateau
20	1:51	ON	22.0	—	—	Plateau (46 s)
21	1:51	ON	22.1	—	—	Plateau breaks
22	1:56	ON	22.2	—	—	Rising
23	1:58	ON	22.3	—	—	= BASELINE
24	2:01	ON	22.4	—	—	First above
25	2:18	ON	24.0	—	—	Linear rise
26	2:30	ON	25.5	—	—	Linear rise
27	2:34	ON	26.4	—	—	Pre-OFF
28	2:34	OFF	26.4	—	—	FLAME OFF
29	6:30	OFF	34.2	43.7	+9.5	Reservoir
30	9:50	OFF	36.0	41.0	+5.0	Reservoir
31	11:03	OFF	36.5	40.8	+4.3	Reservoir
32	11:07	OFF	36.6	26.5*	—	*Wall control
33	12:27	OFF	37.0	40.5	+3.5	Reservoir
34	13:26	OFF	37.3	39.7	+2.4	Reservoir
35	14:44	OFF	37.6	38.7	+1.1	Converging
36	16:25	OFF	38.0	39.0	+1.0	Converging
37	18:12	OFF	38.1	38.5	+0.4	Near convergence
38	20:19	OFF	38.1	38.1	0.0	CONVERGENCE
39	21:23	OFF	38.1	37.7	–0.4	Inverted
40	23:06	OFF	38.1	37.2	–0.9	Inverted
41	24:02	OFF	38.0	37.0	–1.0	Cooling
42	25:57	OFF	37.9	36.5	–1.4	Cooling
43	26:04	ON	37.9	—	—	FLAME ON #2
44	26:25	ON	38.0	—	—	Slow rise

45	26:36	ON	37.9	—	—	Stall
46	26:39	ON	37.8	—	—	Drop (!)
47	26:53	ON	37.8	57.4	+19.6	Surface explosion
48	27:10	OFF	39.0	57.1	+18.1	END

Anomaly Analysis — Experiment 6

Anomaly L1: 46-second plateau at 0.3°C below baseline under maximum flame. From t=1:04 to t=1:51, temperature clamped at 22.0°C (baseline = 22.3°C). 46 seconds of maximum heat input produce zero net temperature increase. The system absorbs all incoming energy endothermically. Recovery to baseline requires 63 seconds of continuous flame (t=0:55 to t=1:58).

Anomaly L2: Temperature rises +11.7°C over 17 min 45 s without any heat source. From flame OFF at 2:34 (probe 26.4°C) to convergence at 20:19 (probe 38.1°C). Classical thermodynamics: temperature should decrease when heat source is removed. Observed: temperature increases continuously. The stored internal energy (endothermic reservoir) is being released as the system equilibrates internally.

Anomaly L3: Surface hotter than bottom for 17+ minutes without flame. From 6:30 (Δ = +9.5°C) to 20:19 (Δ = 0.0°C). The surface is consistently hotter despite no external heat. Classical prediction: without heating from below, the bottom should be warmer (residual heat from pot walls). Observed: surface retains more thermal energy, consistent with the torque-repulsion phase depositing heated matter at the surface during the active heating phase.

Anomaly L4: Second flame cycle — Δ = 19.6°C in 48 seconds. At t=26:53, IR reads 57.4°C while probe reads 37.8°C. The surface absorbs and concentrates thermal energy while the bottom remains cold — the flame attracts cold matter downward (probe measures incoming cold stream), while heated matter is ejected to the surface (IR measures outgoing hot stream). Furthermore, at t=26:39 (33 s of flame), probe reads 37.8°C — a DROP from 37.9°C at flame ignition. Temperature decrease with flame ON, reproducing the Experiment 5 pattern.

Systematic Falsification — Experiment 6

Classical Prediction	Observation	Status
Heating raises temperature	46 s plateau below baseline with max flame	FALSIFIED
T decreases when flame removed	T rises +11.7°C over 17 min without flame	FALSIFIED
Bottom of pot is hottest (heated below)	Surface hotter than bottom for 17+ min	FALSIFIED
T rises monotonically with flame	Probe drops 37.9→37.8 during 2nd flame cycle	FALSIFIED
Surface-bottom gradient requires active heating	Δ=9.5°C persists 4 min after flame OFF	FALSIFIED

Cross-Experiment Unified Analysis

Comparative Parameter Table

Parameter	Exp 1 Iceberg	Exp 2 Butter	Exp 3 Water	Exp 4 Milk	Exp 5 Beans S	Exp 6 Beans L
Material	H■O	Lipid	H■O	Milk	H■O+beans	H■O+beans
T range	0–100°C	25–65°C	72–100°C	69–101°C	15.8–18.8°C	22–39°C
Key finding	Uphill flow	Curved path	27.3°C inv.	0.3°C cycle	–1.5°C drop	17 min rise
Falsification	P1: flows↓	P4: radial	P6: ∝ T	P8: T thresh	P10: T↑	P12: mono↑

Universal Principle Statement

Six materials/configurations — water/ice, lipid/butter, water at boiling, milk emulsion, water with beans+lentils (short and long) — in different phase regimes, different temperature ranges, different gradient magnitudes, different timescales — all produce the same phenomenological signature: **bulk mechanical and thermal behavior is determined by the presence of active thermal flux (flame ON/OFF), not by the absolute temperature of the material.**

Experiments 1–2 demonstrate spatial transport against gravity. Experiments 3–4 demonstrate temporal correlation (motion tracks flame state). Experiments 5–6 demonstrate thermal absorption anomalies (temperature drops with flame, rises without flame) and inverted gradients (bottom colder than top with heating from below).

The thermal force law $F■_{thermal} = -k_T m\nabla T$ unifies all observations. The convective cycle is attraction→torque→repulsion (thermal inverse of magnetic repulsion→attraction). $k_T \approx (4.3–6.4)\times10^{-3}$ $m\cdot s^{-2}\cdot K^{-1}$ (first terrestrial calibration).

Conclusion

Six independent kitchen-scale experiments falsify twelve classical thermodynamic predictions and establish nine new empirical facts. The unified explanation is the thermal force law $F■_{thermal} = -k_T m\nabla T$. The modified Second Law holds that entropy may decrease locally when $|\nabla T| > |\nabla T|_{critical}$ under active flux. The critical distinction established across all six experiments is between stored thermal energy (temperature, scalar, equilibrium property) and active thermal flux (gradient under external forcing, vector, non-equilibrium property). Classical thermodynamics conflates these. This paper distinguishes them empirically.

This framework, combined with the result Gravity = Magnetism + Heat, implies that gravitational attraction contains a thermal component of fundamental significance.

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Supplementary materials: Google Drive · YouTube: @5incere5science · TikTok: @5incere5science

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Experiment 7: The Ice Age — SVE Cryogenic Resilience and Anomalous Volume Contraction

Experimental Overview

This experiment documents three interconnected anomalies related to Structured Vesicular Entities (SVEs) and the freezing behavior of water in different container materials. SVEs are macroscopic structures that form spontaneously in still mineral water — distinct from dissolved gas bubbles — and exhibit properties incompatible with any known gaseous phenomenon. Video: youtube.com/shorts/sDOyiIWXbpU

The experiment is organized into three sub-experiments: (A) SVE formation and comparison with carbonated water, (B) freezing behavior across container materials (glass, copper, plastic), and (C) microscopic characterization of SVEs in liquid water and in ice.

Sub-Experiment A: SVE Formation vs. Gas Loss (C1–C3)

Anomaly Analysis — Experiment 10

Anomaly G1: SVEs exhibit water-equivalent density. During oscillation testing (C1–C2, 12 frames, >330 ms), SVEs follow the exact inertial trajectory of surrounding water with zero measurable phase lag. Classical gas dynamics: a gas bubble ($\rho \approx 1.2$ kg/m^3) in water ($\rho \approx 1000$ kg/m^3) experiences net upward buoyancy $F_b = (\rho_{water} - \rho_{gas})Vg \approx \rho_{water}Vg$. During downward water acceleration, gas would lag upward relative to water; during upward acceleration, gas would lag downward. The 833× density ratio produces observable separation within milliseconds of oscillation onset. No separation is observed. The density of SVEs equals the density of water to within the resolution of the measurement (~1%).

Anomaly G2: SVEs persist at depth while confirmed gas rises. The air injection control (C3, F3–F6) introduces a direct comparison: atmospheric air injected via syringe into the same liquid, at the same moment, alongside SVEs already present. The injected air rises to the surface within seconds. The SVEs remain at depth, stationary. This eliminates all gas-phase hypotheses in a single observation: if SVEs were gas of any composition (CO_2, N_2, O_2, or any mixture), they would rise. They do not. The only interpretation consistent with stationary suspension at depth is $\rho_{SVE} \approx \rho_{water}$.

Anomaly G3: SVEs survive CO_2 degassing. After vigorous shaking (C3, F1), both gas bubbles and SVEs are visible. After rest (C3, F2), the gas has risen and dissipated — as gas does. The SVEs remain. This temporal persistence eliminates the hypothesis that SVEs are transient gas nucleation sites: the gas has left, but the structures persist. They are materially distinct from the dissolved gas that was temporarily released by shaking.

Systematic Falsification — Experiment 10

Classical Prediction	Observation	Status
Gas bubbles in liquid exhibit buoyancy lag during oscillation ($\rho_{gas} \blacksquare \rho_{liquid}$)	SVEs follow exact water inertia with zero phase lag across 12 frames	**FALSIFIED**
Gas bubbles rise to surface in liquid at rest	SVEs remain stationary at depth while injected air rises instantly	**FALSIFIED**
Dissolved CO_2 structures dissipate upon degassing	SVEs persist after all shaking-released gas has risen and dissipated	**FALSIFIED**
Visible structures in mineral water are gas phase	SVEs have water-equivalent density and behave as water-density structures	**FALSIFIED**

Theoretical Significance

This experiment addresses the most common dismissal of SVE observations: "those are just gas bubbles." The density equivalence demonstrated here is categorically incompatible with any gas-phase interpretation. Gas at atmospheric pressure has density ~1.2 kg/m^3; water has density ~1000 kg/m^3. Even compressed CO_2 at the pressures present in mineral water (~3–4 atm) has density ~6–8 kg/m^3 — still >100× less dense than water. No gas at any pressure achievable in a sealed commercial bottle approaches water density.

The oscillation test provides a direct density measurement: objects with different density than their surrounding fluid exhibit buoyancy-driven phase lag during oscillation. This is the operational principle behind settling columns, centrifuges, and inertial separation. SVEs show no such lag. Their density equals that of water.

The air injection control (C3) elevates this from a negative result (absence of lag) to a positive demonstration (simultaneous comparison): in the same container, at the same moment, gas rises and SVEs do not. The falsification is not statistical — it is categorical. One population behaves as gas; the other does not.

Combined with previous experiments documenting SVE internal structure (concentric layers, fracture surfaces, material integrity under 1000× microscopy — Experiments 7–8 in the Ice Age series), the present result completes the phenomenological characterization: SVEs are water-density structures with internal organization, not gas bubbles with a simple gas-liquid interface.

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1. Clausius, R. (1865). The Mechanical Theory of Heat. London: John van Voorst.
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